

5G

Module 44

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Video: <https://www.youtube.com/watch?v=vtgDWzFM8Uo>

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0:00 hello viewers I welcome you to the series of lectures on cyber security and information security today we will

0:08 discuss 5G what is 5G fifth generation mobile networks or fifth generation Wireless Systems is

0:17 abbreviated as 5G it is used to describe the upcoming fifth generation of mobile

0:24 network technology it is the coming fifth generation Wireless Broadband technology which is based on the i e

0:33 802.11 AC standard 5G technology is going to be a new mobile revolution in Mobile Market

0:41 through 5G technology now we can use Worldwide Cellular phones with the coming of cell phone alike to PDA now

0:49 your whole office is on your fingertips or in your phone 5G technology has extraordinary

0:56 data capabilities and has ability to tie together unrestricted call volumes and infinite data broadcast within latest

1:05 mobile operating system the following figure shows the speed of 3G 4G and 5G

1:13 in the form of a dashboard meters what exactly is 5G the G in 5G stands for

1:21 Generation the wireless phone technology technically started with 1G in the early

1:27 1990s and 1G expanded to 2G when people are able to send text messages between two cellular devices and this is made by

1:36 only the companies after this the world moved towards 3G with this people were capable

1:44 to make phone calls send text messages and browse the internet four generation enhanced many of the capabilities that

1:53 were made possible with the third generation of Wireless people could browse the web send text messages and make phone calls

2:02 and they could even download and upload large video files without any issues after 4G companies added LTE that

2:12 is long-term Evolution with 4G connectivity LTE became the fastest and

2:19 most consistent variety of 4G compared to competing technologies like ymax the
difference between ymax and LT

2:28 is both Technologies achieved similar outcomes but it was important to create a
standard for everyone to

2:37 use LT makes the 4G technology faster 5G will provide better speeds and

2:45 coverage than the 4G 5G works with a 5 GHz signal and is set to offer speeds of

2:52 up to 1 gabit per second for tens of connections or tens of megabits per

2:59 second for tens of thousands of connections 5G will build on the foundation created
by 4G

3:07 LTE it's going to allow people send texts make calls and browse the web as

3:13 always but with the increase of speed 5D will make easy for people to download

3:19 and upload Ultra HD and 3D videos the Next Generation mobile networks Alliance
defines the following

3:28 requirements start of 5G standard should fulfill data rates of tens of megabits

3:35 per second for tens of thousands of users data rates of 100 megabits per second for
metropolitan

3:44 areas 1 gab per second simultaneously to many workers on the same office

3:51 floor several hundreds of thousands of simultaneous connections for massive
wireless sensor Network

3:59 spectral efficiency is significantly enhanced compared to 4G coverage

4:06 improved signaling efficiency enhanced 1 millisecond latency or very

4:13 less we can say virtually zero latency smart communication latency reduced

4:20 significantly compared to LTE so some of these features can be shown in this
diagram so now we come to design of of

4:29 5G mobile network architecture this is a model that proposes design of network
architecture

4:37 for 5G mobile systems is all IP based model for wireless and mobile networks

4:45 interoperability the system consists of a user terminal which has a crucial role in a
new architecture and a number of

4:53 independent autonomous radio access Technologies within each of the terminals
each of of the radio access

5:00 Technologies is seen as the IP Link to the outside internet World however there
should be different radio interface for

5:09 each radio access technology R in the mobile terminal for an example if we

5:16 want to have access to four different RS we need to have four different exess
specific interfaces in the mobile

5:24 terminal and to have all of them active at the same time with aim to have this
architecture to be

5:32 functional is 5G really faster than 4G yes 5G is really much faster than 4G
5:40 and their speeds will be significantly faster now times 4G LTE transfer speeds
5:47 top out at about 1 gbit per second this means it takes about an hour to download
5:54 a short HD movie in perfect conditions but the problem is that people rarely
experience 4g's maximum download speed
6:03 because of the signal can be disrupted by so many different things for example
buildings microwaves other Wi-Fi signals
6:12 and many more 5G will increase download speeds up to 10 gbits per second this
means a full
6:20 HD movie can be downloaded in seconds it will also reduce latency significantly we
can say that it
6:28 provides the Pass 2 thousands of connected devices that are placed at our homes
and offices concepts for 5G mobile
6:38 networks the 5G terminals will have softwar defined radios and modulation
schemes as well as new error control
6:47 schemes that can be downloaded from the internet the development is seen
towards the user terminals as a focus of the 5G
6:56 mobile networks the terminals will have access to different Wireless Technologies
at the same time and the terminal should be
7:05 able to combine different flows from different Technologies the vertical handovers
should be avoided because they
7:12 are not feasible in case when there are many Technologies and many operators and
service providers in 5G each network
7:21 will be responsible for handling user Mobility while the terminal will make the final
Choice among different
7:28 wireless mobile file Access Network providers for a given Service Such choice will
be based on an open
7:36 intelligent middleware in the mobile phone the system comprising of a main user
terminal and then a number of
7:45 independent and autonomous radio access Technologies each of the radio
Technologies is considered as the IP
7:52 Link for the outside internet world how does it work there there are already huge
group
8:00 of major Global telecoms working to create worldwide standards around 5G as
8:07 we already have 4G and 3G so experts think or expects it is compatible with them
but till now most of the standards
8:15 are not solidified as in the most basic form cell phones are basically two way radios
8:23 when we call someone then our phone converts our voice into an electrical signal it
transmits that electrical

8:31 signal to the nearest Cellar using radio wave The Cellar bounces the radio wave
through a network of SARS and then to

8:40 our friends phone the same is happening when we send other forms of data like
photos and video across the

8:49 network as when a new mobile wireless technology comes along like 5G it is
8:56 assigned a higher radio frequency 4G occupied the frequency bands up to 20
9:02 MHz but in the case of 5G it will likely to use the frequency band up to 6
9:09 GHz the reason new wireless Technologies occupy higher frequencies is because
they typically aren't in use and move

9:17 information at a much faster speed the problem is that higher frequency signals
don't travel as far as lower frequencies

9:26 so multiple input and output in will probably be used to boost signals anywhere 5G
is

9:35 offered 5G is an opportunity to develop a more sustainable operator investment
model if previous generations of mobile

9:45 technology have taught us anything it is that as with each preceding generation 5G
will unlock value in ways we cannot

9:54 and will not anticipate services that were initially expected to have a negligible
impact

10:01 10 minutes, 1 second became hugely popular example SMS while those expected to
be next big thing that

10:08 have slow to gain transaction example video calling with the development of 5G the
10:16 industry can expect a paradigm shift in the way that all the stakeholders in the
mobile ecosystem play their

10:24 role Regulators especially can have this opportunity to create healthier
environments that stimulate continuing

10:32 investment in Next Generation technology some of the business cases that have
worked well for 3G and 4G

10:41 Technologies may not be right ones for 5G by continuing work operators have the
10:47 new paradigm difference between 5G and 4G 4G networks are being widely
deployed

10:55 around the world and although the technology continues to evolve the focus now
switches to 5G the fifth generation

11:03 of mobile networks dubbed 5G the Next Generation Network may still be several
years away from being deployed but it is

11:12 the buzz World in telecommunications at the moment 5G is the next major phase of
11:19 mobile telecommunication and follows the global adoption of 4G LTE that is
11:26 long-term Evolution standards since 2012 a new telecommunication standard is
11:32 introduced approximately every 10 years 2G was introduced in

11:38 1991 3G in 2001 and 4G in 2012 and 5G is
11:45 expected to be made available in 2020 or 2022 the fifth generation of mobile
11:52 technology does not have a defined universal standard although a few
new likely Technologies are
12:00 emerging and whereas 4G connections offer average speeds of 2540 Mbps at
12:08 home 5G is likely to offer speeds in excess of 500 Mbps many entities believe
12:16 that 5G will be defined by its ability to offer 1 gbps per second mobile speeds to
10 simultaneously connections
12:26 and Chinese mobile Giant huai expects 5G to be at least 100 times faster than the
12:34 fastest 4G LTE standard currently available a 4G plus speed test 5G could
12:42 be up to 100 times faster 5G Technologies and
12:49 frequencies 4G in the UK currently uses the LTE standard with radio waves
operating on the 800
12:58 mhz 1800 mhz and 2.6 GHz frequency bands the 1800 mhz band were
repurposed
13:08 by E and three to use with LTE while as the other frequencies were made available
to all networks in the 4G
13:17 Spectrum auction back in 2012 who I believe that 5G will be based
13:24 upon a combination of both new radio access Technologies are and existing
Wireless Technologies such
13:32 as LTE HSPA GSM and Wi-Fi whilst others such as Samsung and researchers at
New
13:40 York University have discussed the concept of using millimeter wave frequencies
millimeter wave frequencies
13:49 lie between 30 to 300 mhz which is much higher than existing Network standards
13:56 and offer the key benefit that are scarcely used by other broadcast Technologies
such as AM FM radio and TV
14:05 which means it has the potential for greater speeds and the capability of more data
to pass through it the
14:13 millimeter wave approach does not have some potential downsides as these waves do
not pass through solid objects very
14:22 well which makes them difficult to sustain over long distances as a result mm
wave
14:30 approach would use lots of smaller base stations whereas current mobile networks
rely on a combination of large base
14:39 stations and smaller ones in densely populated areas and it's expected that
14:45 this approach will be used by all 5G standards to ensure maximum capacity and
minimal
14:53 latency as mentioned elsewhere there is no Universal 5G standard defined and

when eventually launched we could see a

15:02 variety of Standards operating under the 5G Banner similar to how both LTE and
15:08 yaax and American LTE standard used by CDMA networks operate under the 4G
15:15 Banner what are the benefits of 5G the potential benefits to 5G are vast
15:23 for both businesses and consumers for the former the additional capacity and
speed would allow for Greater Mobile
15:31 working Wireless for consumers the speed should offer additional benefits within
the ability of your
15:39 smartphone example a business using a 5G Network would mean employees can
video conference from any location whilest for
15:48 consumers 5G could allow you download a film to your smartphone in under a
second aside from the greater speed
15:57 another key benefit to 5G is the greater capacity current 4G contracts offer
16:03 Promises of speeds up to 150 megabits per second on the go but leave must users
with inclusive data allowances of
16:12 less than 5G although most networks do offer plans with higher alliances at an
extra charge due to the limited
16:21 capacity 5G is expected to offer upwards of 1,000 times the capacity of 4G which
16:28 would allow mobile networks to offer greater data allowances as part of their
Mobile plans 4G speed test showing
16:37 latency of 49 milliseconds another widely agreed benefit of 5G regardless of
standard
16:44 adopted is reduced latency latency is defined as the amount of time taken for data
transfer to take
16:53 place example the buffering period when streaming a video or the loading period
when opening a web page and as the
17:02 number of users on 4G has increased so has the latency with 5G huai is targeting
17:10 latency of less than 1 millisecond compared to latency in excess of 50 millisecond
on 4G which should mean that
17:19 your video starts playing as soon as you hit play the reduced latency will also have
17:26 wide reaching benefits in video games automated equipment The Internet of Things
Arena and augmented reality for
17:35 the later especially reduced latency in 5G could deliver exceptional experiences
without the delays currently faced when
17:44 using augmented reality whether it's the Oculus Rift
17:50 gear VR or Microsoft Hol lens all of these can benefit from reduced latency
17:58 in order to deliver the most realistic experience a likely additional benefit of 5G
depending on the standard used is
18:07 that it will allow current network coverage to expanded into hard to reach areas

while West 4G is widely being

18:16 rolled out across the UK there are areas where coverage remains minimal at best and with 5G networks will be able to

18:25 reach further areas by deploying multiple smaller antennas which will allow signals to be emitted in multiple

18:32 directions and even bounced off Solid Surfaces now what are the requirements of

18:39 5G 5G has better coverage area and high data rate at the edge of the cell it has

18:47 low battery consumption so battery life will be much longer availability of multiple data

18:54 transfer path around 1 Gbps per second data rate is easily possible security is

19:03 more Energy Efficiency and spectral efficiency are good worldwide coverage

19:10 about 90% reduction in network energy usage because of the advantages fifth

19:18 generation wireless system is very much Essential now we come to the applications of 5G

19:25 network network availability will be everywhere and will facilitate people to use their computer and such kind of mobile devices

19:34 anywhere anytime it will make the world real Wi-Fi zone it allow people to use their mobile

19:43 and laptops and other devices anywhere and at any time as network will be available

19:50 everywhere mobile IP address will be assigned to the devices depends on the connected Network and geographical

19:57 position by using IP V6 addressing technology it will make unified Global

20:05 standards for all by using 5G technology we can receive radio signals even at a higher

20:13 altitude now we come to the disadvantages of 5G technology now as we know 5G technology

20:20 is researched and conceptualized to solve all radio signal problem but because of some technical advancements

20:28 of some of the geographical areas and some of the security reasons it has the following

20:35 disadvantages research is still going on it and this is under process the speed claims this technology

20:43 seems difficult to achieve as there are other devices and incompetent technology involved in other parts of the

20:55 world many of the old devices to be replaced as these are not competent to 5G which increases the

21:03 cost developing infrastructure will also increase the cost security and private issues yet to

- 21:11 be solved so now what is the future span of 5G the future enhancement of Nano core
- 21:18 will be incredible as it combines with artificial intelligent one can able to control his intelligent robot using his
- 21:27 mobile phone our mobile can automatically type the message what our brain thingss the Google hot Trends have rated
- 21:36 the term 6G as the 17th most searched word in the search engines the iPod 6G comes in seven
- 21:45 different colors and has a aluminium body which makes the body strong to withstand constant daily
- 21:52 usage so that's all for the 5G technology thank you viewers for watching this program